# Object detection

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Use Python to demonstrate **Ultralytics**: the effect of Object Detection on the images, the videos and the real time detection.

## 1. Enable the maximum performance of the mainboard

### 1.1. Enable the MAX power mode

Enabling the MAX Power Mode (the maximum power mode) on the Jetson ensures that all the CPU and GPU cores are turned on:

```
sudo nvpmodel -m 0
```

### 1.2. Enable the Jetson clocks

Enabling the Jetson Clocks ensures that all the CPU and GPU cores run at the maximum frequency:

```
sudo jetson_clocks
```

## 2. Object prediction: the image

Use yolo26n.pt to predict the images that come with the ultralytics project.

Enter the code folder:

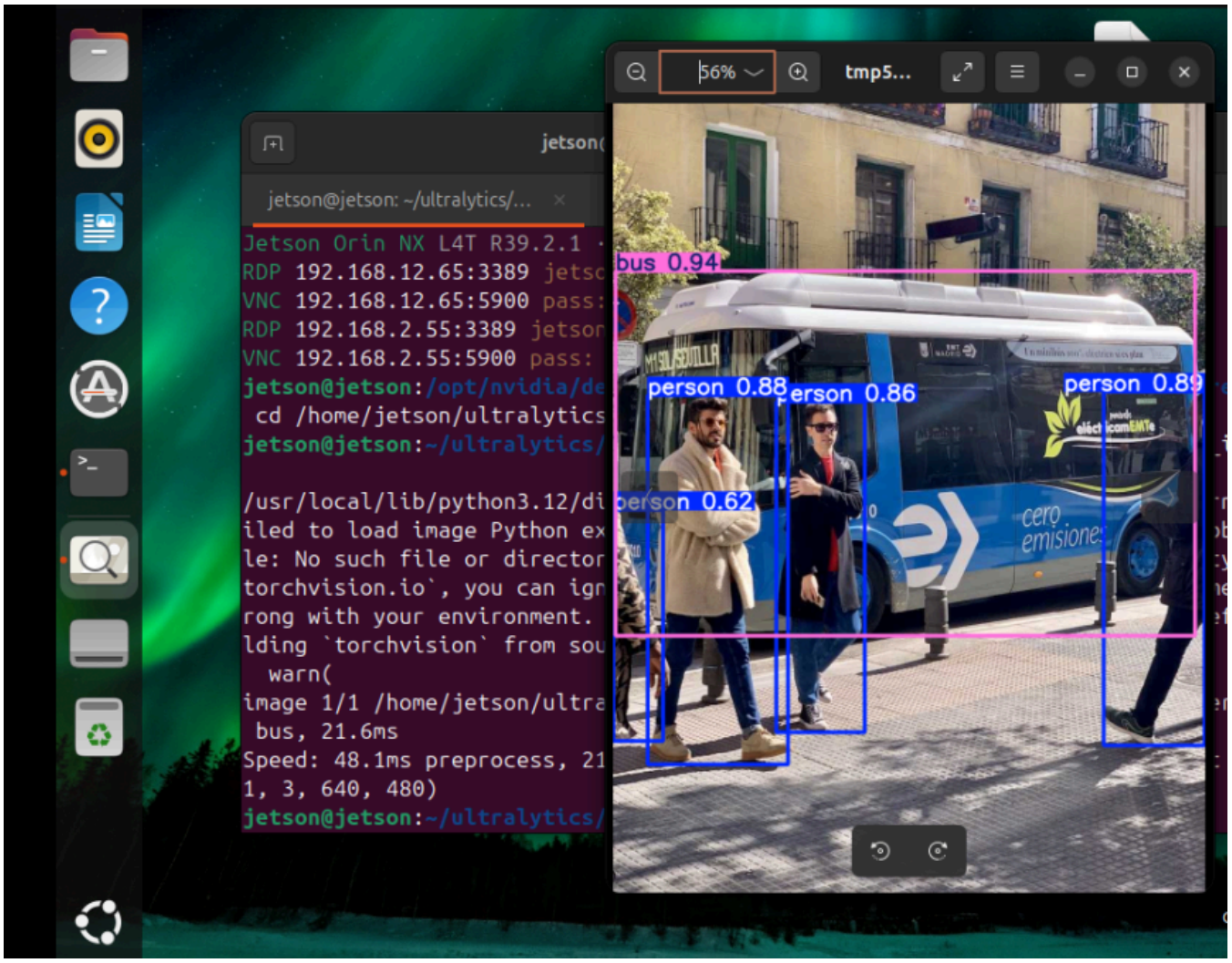
```
cd /home/jetson/ultralytics/ultralytics/yahboom_yolo26
```

Run the code:

```
python3 01.detection_image.py
```

## Effect preview

The location of the image output by yolo: /home/jetson/ultralytics/ultralytics/output/




The sample code:

```python
from ultralytics import YOLO

# Load a model
model = YOLO("/home/jetson/ultralytics/ultralytics/yolo26n.pt")

# Run batched inference on a list of images
results = model("/home/jetson/ultralytics/ultralytics/assets/bus.jpg")  # return a list of Results objects

# Process results list
for result in results:
    boxes = result.boxes  # Boxes object for bounding box outputs
    # masks = result.masks  # Masks object for segmentation masks outputs
    # keypoints = result.keypoints  # Keypoints object for pose outputs
    # probs = result.probs  # Probs object for classification outputs
    # obb = result.obb  # Oriented boxes object for OBB outputs
    result.show()  # display to screen
```

```
    result.save(filename="/home/jetson/ultralytics/ultralytics/output/bus_output.jpg")  # save to disk
```

# 3. Object prediction: the video

Use yolo26n.pt to predict the videos under the ultralytics project (not the videos that come with ultralytics).

Enter the code folder:

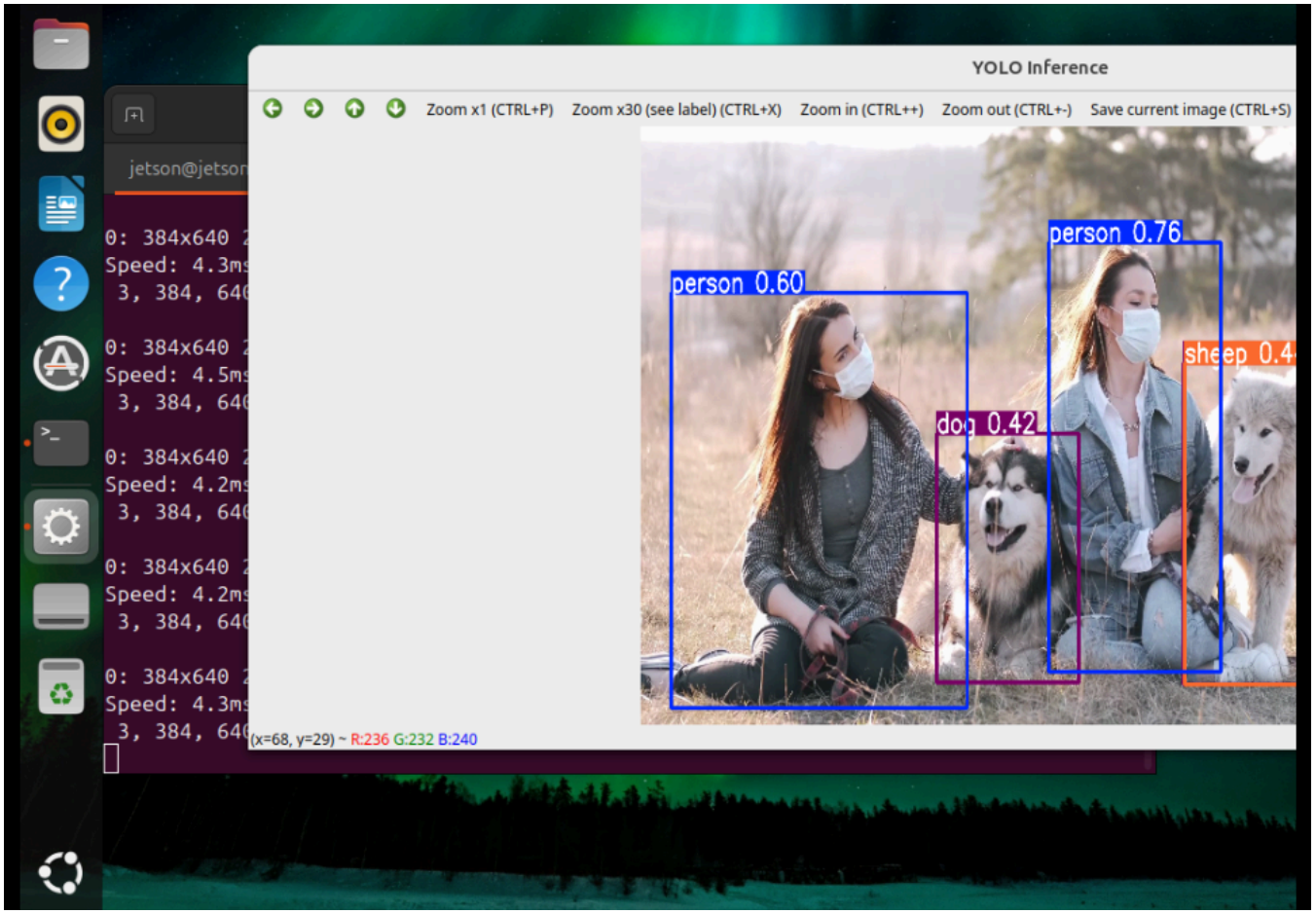
```
cd /home/jetson/ultralytics/ultralytics/yahboom_yolo26
```

Run the code:

```
python3 01.detection_video.py
```

## Effect preview

The location of the video output by yolo: /home/jetson/ultralytics/ultralytics/output/




The sample code:

```
import cv2
from ultralytics import YOLO

# Load the YOLO model
model = YOLO("/home/jetson/ultralytics/ultralytics/yolo26n.pt")
```

```
# Open the video file
video_path = "/home/jetson/ultralytics/ultralytics/videos/people_animals.mp4"
cap = cv2.VideoCapture(video_path)

# Get the video frame size and frame rate
frame_width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
frame_height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = int(cap.get(cv2.CAP_PROP_FPS))

# Define the codec and create a VideoWriter object to output the processed video
output_path = "/home/jetson/ultralytics/ultralytics/output/01.people_animals_output.mp4"
fourcc = cv2.VideoWriter_fourcc(*'mp4v')  # You can use 'XVID' or 'mp4v' depending on your
platform
out = cv2.VideoWriter(output_path, fourcc, fps, (frame_width, frame_height))

# Loop through the video frames
while cap.isOpened():
    # Read a frame from the video
    success, frame = cap.read()

    if success:
        # Run YOLO inference on the frame
        results = model(frame)

        # Visualize the results on the frame
        annotated_frame = results[0].plot()

        # Write the annotated frame to the output video file
        out.write(annotated_frame)

        # Display the annotated frame
        cv2.imshow("YOLO Inference", annotated_frame)

        # Break the loop if 'q' is pressed
        if cv2.waitKey(1) & 0xFF == ord("q"):
            break
    else:
        # Break the loop if the end of the video is reached
        break
# Release the video capture and writer objects, and close the display window
cap.release()
out.release()
cv2.destroyAllWindows()
```

# 4. Object prediction: the real time detection

## 4.1. USB camera

Use yolo26n.pt to predict the image of the USB camera.

Enter the code folder:

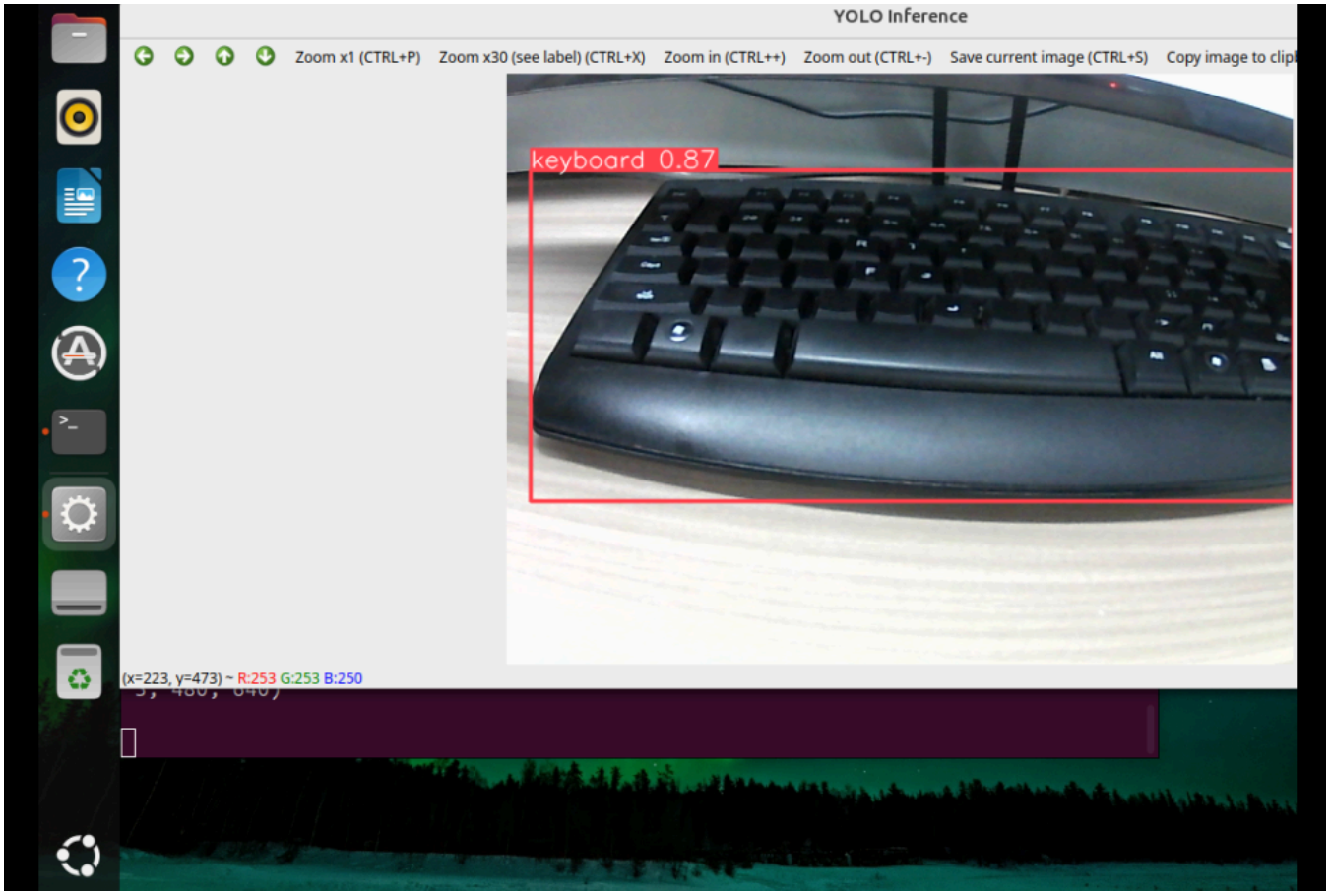
```
cd /home/jetson/ultralytics/ultralytics/yahboom_yolo26
```

Run the code: click the preview window and press the q key to terminate the program!

```
python3 01.detection_camera_usb.py
```

## Effect preview

The location of the video output by yolo: /home/jetson/ultralytics/ultralytics/output/




The sample code:

```
import cv2
from ultralytics import YOLO

# Load the YOLO model
model = YOLO("/home/jetson/ultralytics/ultralytics/yolo26n.pt")

# Open the cammera
cap = cv2.VideoCapture(0)

# Get the video frame size and frame rate
frame_width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
frame_height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = int(cap.get(cv2.CAP_PROP_FPS))

# Define the codec and create a VideoWriter object to output the processed video
```

```
output_path = "/home/jetson/ultralytics/ultralytics/output/01.detection_camera_usb.mp4"
fourcc = cv2.VideoWriter_fourcc(*'mp4v')  # You can use 'XVID' or 'mp4v' depending on your platform
out = cv2.VideoWriter(output_path, fourcc, fps, (frame_width, frame_height))

# Loop through the video frames
while cap.isOpened():
    # Read a frame from the video
    success, frame = cap.read()

    if success:
        # Run YOLO inference on the frame
        results = model(frame)

        # Visualize the results on the frame
        annotated_frame = results[0].plot()

        # Write the annotated frame to the output video file
        out.write(annotated_frame)

        # Display the annotated frame
        cv2.imshow("YOLO Inference", annotated_frame)

        # Break the loop if 'q' is pressed
        if cv2.waitKey(1) & 0xFF == ord("q"):
            break
    else:
        # Break the loop if the end of the video is reached
        break
# Release the video capture and writer objects, and close the display window
cap.release()
out.release()
cv2.destroyAllWindows()
```

## References

https://docs.ultralytics.com/tasks/detect/